When Does Test-Time Ensembling Help Large Language Models? A Budget-Matched Study of Answer-Space and Prompt-Space Diversity

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Abstract

Test-time ensembling improves the reliability of large language models by aggregating answers from multiple model calls instead of trusting a single greedy prediction. Two families of strategies dominate practice. Self-consistency resamples the stochastic decoder on one fixed prompt, so the ensemble varies along what we call the answer space. Prompt ensembling instead fixes the decoding configuration and varies the prompt over semantically equivalent paraphrases, so the ensemble varies along the prompt space. Both families consume the same scarce resource, namely the number of model calls, yet prior work studies them in isolation and the question of which source of diversity buys more accuracy per call remains open. We present a controlled, budget-matched comparison across six commercial large language models and six reasoning and knowledge benchmarks under a unified protocol in which every question is answered sixteen times per model: six decodes of the original prompt, six paraphrased prompts answered once each, a hybrid condition that combines three paraphrases with two decodes each, a twelve-vote combined condition, and a greedy baseline. The results reveal a regime structure that current evaluation practice overlooks. On benchmarks that modern flagship models saturate, the sampled answer distribution degenerates to a single value, both ensembles collapse onto the greedy baseline, and consistency-based uncertainty estimates lose their discriminative power, which raises a caution for hallucination detection methods built on sampling.

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In regimes with genuine answer entropy, both ensembles lift accuracy over greedy decoding, the two diversity sources are statistically indistinguishable at matched budgets while disagreeing on individual questions, majority voting can even reduce accuracy when the modal answer is wrong, and the two agreement signals detect complementary errors across models. We distill the findings into a practical decision rule for allocating a fixed test-time budget, and we release all prompts, paraphrases, and raw predictions to support reproducible evaluation.

Keywords: Large language models, test-time compute, self-consistency, prompt ensembling, uncertainty estimation, robustness

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1. Introduction

Large language models answer questions through a single forward pass whose output depends on two design choices that the user controls: the prompt that specifies the task, and the decoding configuration that maps model probabilities into text. Both choices turn out to be fragile. A long line of work on prompt sensitivity shows that semantically equivalent reformulations of the same question can change model answers by large margins [1, 2, 3, 4]. A parallel line of work on test-time compute shows that a single greedy decode is an unreliable summary of what the model knows, and that aggregating several decodes of the same prompt, as in self-consistency [5], improves reasoning accuracy at the cost of additional inference calls.

These two observations motivate two different ways of spending a fixed budget of model calls. The first spends the budget on the answer space: keep the prompt fixed, sample several decodes at nonzero temperature, and aggregate them by majority vote. The second spends the budget on the prompt space: keep decoding deterministic or lightly stochastic, present several semantically equivalent paraphrases of the task, and aggregate the resulting answers. Both strategies are used in practice, both appear in the evaluation literature under separate headings, and both consume the same scarce resource. Surprisingly, the literature offers little guidance on which source of diversity yields more accuracy per call, whether the two sources detect the same errors, and how the answer changes with model capability and task difficulty.

The question is becoming more urgent as model behavior shifts. Post-trained flagship models are increasingly confident: on many legacy bench-

marks they answer near ceiling, and preliminary reports suggest that repeated sampling at moderate temperature often returns identical answers, which would render answer-space ensembling inert precisely where deployments are most common. At the same time, uncertainty estimation methods built on sampling, such as semantic entropy [6, 7], implicitly assume that the sample distribution carries information about correctness. If that assumption fails for confident models, practitioners need to know before they build on it.

This paper presents, to our knowledge, the first budget-matched, within-question comparison of answer-space and prompt-space ensembling across multiple model families and task types. Every question in our protocol is answered sixteen times by every model under matched conditions: one greedy decode, six stochastic decodes of the original prompt, one stochastic decode of each of six validated paraphrases, and three additional second decodes that support a hybrid condition. From this single call matrix we derive self-consistency ensembles of any size up to six, paraphrase ensembles of the same sizes, a hybrid allocation, a combined twelve-vote ensemble, per-question consistency scores for both diversity sources, and fragility measures that record how often paraphrasing alone flips the majority answer. Because all conditions are derived from the same matrix, comparisons between strategies hold the question, the model, and the total call budget constant by construction.

Our study addresses four research questions. First, under equal budgets, does spending calls on paraphrases buy more, equal, or less accuracy than spending them on resampling, and does the hybrid allocation dominate either pure strategy? Second, how brittle are model answers to paraphrasing per se, and where do paraphrase-induced flips concentrate? Third, can agreement across paraphrases serve as an error signal in the way that agreement across samples does, and how do the two signals compare when both are available? Fourth, how do all of these effects interact with model capability and task difficulty? The last question turns out to govern the first three.

The empirical picture that emerges has a clear regime structure. When a model finds a task easy, its answer distribution collapses: across our six benchmarks, flagship models return the same answer for essentially every sample and every paraphrase on saturated tasks, self-consistency and paraphrase ensembling both collapse onto the greedy baseline, and consistency scores carry no signal about which answers are wrong. When a task genuinely challenges a model, answer entropy reappears, both ensemble families improve over greedy decoding, the two consistency measures detect partially

disjoint errors, and a hybrid allocation protects against a failing diversity source. This structure implies that no single published claim about ensembling generalizes across regimes, and that practical systems should first measure the cheap statistics that identify the regime before choosing how to spend their budget.

Our contributions are as follows. We formalize test-time ensembling as a choice between two diversity sources under a shared budget, and we introduce a call matrix design that derives all conditions of interest from one matched set of calls. We provide the first multi-model, multi-benchmark empirical map of when each source helps, including the first systematic evidence that paraphrase agreement functions as an uncertainty signal complementary to sample agreement. We document the answer degeneracy phenomenon that renders both ensembling and consistency-based uncertainty estimation inert on saturated tasks for confident models, with direct implications for hallucination detection methods built on sampling. Finally, we translate the findings into an actionable allocation rule and release our complete prompt materials, validated paraphrases, and per-call predictions.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the ensembling framework and the protocol. Section 4 describes models, benchmarks, and validation procedures. Section 5 presents the results around our four research questions. Section 6 discusses implications and limitations, and Section 7 concludes.

2. Related Work

2.1. Test-time compute and self-consistency

Scaling computation at inference time has become a standard tool for improving the reliability of large language models. Chain-of-thought prompting demonstrates that eliciting intermediate reasoning steps raises accuracy on arithmetic and symbolic tasks [8], and self-consistency extends this observation by sampling multiple reasoning paths at nonzero temperature and returning the most frequent final answer [5]. Follow-up work reduces the number of samples needed through early stopping [9], generalizes aggregation to free-form generation by letting the model itself select among candidate answers [10], and studies the optimal allocation of test-time compute relative to model size [11]. Process reward models that score intermediate steps represent a further refinement of the same budget idea [12]. Multi-agent debate aggregates answers across multiple model instances and rounds

[13], and controlled comparisons find that debate does not reliably outperform self-consistency at matched budgets [14], which reinforces the need for budget-matched comparisons when evaluating aggregation strategies. Our work differs from this literature in the source of diversity it studies: instead of asking how many samples to draw or how to weight them, we ask whether the same calls are better spent perturbing the prompt, and we answer under a matched-budget protocol.

2.2. Prompt sensitivity and prompt ensembling

A separate literature documents that model outputs are sensitive to semantically irrelevant details of the prompt. Reformatting alone, such as changing spacing or option ordering, shifts accuracy by several points [2], and the measured performance of a model depends strongly on which of many equivalent prompts an evaluator happens to choose [1]. Linguistic properties of the prompt, such as lexical familiarity, partially predict its success [3], and adversarial prompt perturbations can degrade performance further [4]. The natural constructive response is to aggregate over prompts. Ask Me Anything generates many question reformulations with a language model and combines the resulting pseudo-labels [15], an idea that predates and parallels self-consistency but targets zero-shot classification rather than matched-budget reasoning. Evaluation studies propose averaging over prompt templates to stabilize benchmark scores [1]. What remains missing is a direct comparison that holds the budget, the model, and the question fixed and contrasts prompt-space aggregation with answer-space aggregation on the same instances, which is the gap our protocol fills.

2.3. Uncertainty estimation from sampled behavior

Between the two aggregation literatures sits a third that uses sampling not to improve accuracy but to estimate uncertainty. Verbalized confidence asks the model to state a probability [16, 17], and self-probing methods ask the model to evaluate its own candidate answers [18]. Behavioral methods infer uncertainty from the dispersion of sampled outputs, from simple answer agreement to semantic clustering that treats paraphrased answers as equivalent [6, 7], with surveys consolidating the rapidly growing set of measures [19]. Calibration metrics such as expected calibration error [20] and selective prediction metrics such as the area under the risk-coverage curve quantify how well a confidence score ranks correct above incorrect answers. These methods inherit an assumption from sampling: that the sample distribution

is informative. Our study tests this assumption directly, characterizes the regimes in which sample agreement and paraphrase agreement carry signal, and shows that the two agreement signals are complementary where both vary. In this sense our fragility analysis also connects to work on the faithfulness of chain-of-thought explanations, which finds that stated reasoning does not always cause the final answer [21]: paraphrase stability measures the robustness of the answer to surface form rather than to the reasoning trace.

2.4. Positioning

In summary, answer-space ensembling, prompt-space ensembling, and sampling-based uncertainty estimation have developed largely in isolation. Answer-space methods assume resampling generates useful diversity. Prompt-space methods assume paraphrases probe the same knowledge. Uncertainty methods assume sample dispersion tracks correctness. All three assumptions are regime dependent, and no prior work, to our knowledge, measures all three under a single matched-budget protocol on modern models. The comparison also carries practical weight for evaluation practice, since benchmark conclusions about model capability shift when prompts vary [1] and when aggregation is applied [5]. Our contribution is the unified empirical map, the validation machinery for paraphrases, and the decision guidance that follows from the regime structure.

3. A Budget-Matched Framework for Test-Time Ensembling

3.1. Setup and notation

Let M denote a large language model and let q denote a question with a gold answer $y^*(q)$. For multiple-choice tasks the answer is an option letter, and for free-form numeric tasks the answer is a value compared numerically. A prompt x is a rendering of q into natural language, consisting of an optional passage, a question stem, an option list, and an instruction that specifies the required output format. We distinguish one canonical prompt x_0 and K paraphrased prompts $x_1, \ldots, x_K$ that preserve the semantics of x_0 under a validation procedure described in Section 4.

The model produces an answer through stochastic decoding. We write $a(M, x, s)$ for the answer extracted from the s-th independent decode of prompt x at temperature $\tau = 0.7$, and $a_g(M, x)$ for the greedy decode at $\tau = 0$. Answers are extracted from the model output by a fixed parser that

searches for the mandated answer format and falls back to pattern matching over the final output lines; the parse rate is reported in Section 5.

3.2. The call matrix

For every question and model we collect a matrix of sixteen calls:

$$\mathcal{C}(M,q) \;=\; \big\{a_g(M,x_0)\big\} \;\cup\; \bigcup_{s=1}^{S}\big\{a(M,x_0,s)\big\} \;\cup\; \bigcup_{k=1}^{K}\big\{a(M,x_k,s) : s \le r_k\big\}, \tag{1}$$

with $S = 6$ samples on the canonical prompt, $K = 6$ paraphrases, and $r_k = 2$ decodes for the first three paraphrases and $r_k = 1$ for the remaining three. The matrix therefore contains one greedy call, six canonical samples, six first-decode paraphrase calls, and three second-decode paraphrase calls. Every experimental condition studied in this paper is a deterministic function of this matrix, which guarantees that all comparisons hold the question, the model, and the realized decoding randomness constant across conditions.

3.3. Ensemble conditions

All conditions aggregate votes by majority rule. Ties are broken in favor of the earliest vote in a fixed canonical order, which prefers the canonical prompt over paraphrases and earlier paraphrases over later ones; we report tie incidence, which is negligible except where noted.

The *self-consistency* condition of size k aggregates the first k canonical samples:

$$\text{SC-}k(q) \;=\; \text{majority}\big(a(M,x_0,1),\ldots,a(M,x_0,k)\big), \qquad k \le 6. \tag{2}$$

The *paraphrase ensemble* of size k aggregates one decode of each of the first k paraphrases:

$$\text{PE-}k(q) \;=\; \text{majority}\big(a(M,x_1,1),\ldots,a(M,x_k,1)\big), \qquad k \le 6. \tag{3}$$

Both conditions spend exactly k model calls and differ only in where the diversity comes from: SC-k perturbs the decoder while the prompt is fixed, whereas PE-k perturbs the prompt while the decoder configuration is fixed. This is the budget match that makes the two conditions comparable.

The *hybrid* condition splits a budget of six calls between the two sources, using two decodes of each of the first three paraphrases:

$$\text{HB-6}(q) \;=\; \text{majority}\big(a(M,x_k,s) : k \in \{1,2,3\},\ s \in \{1,2\}\big). \tag{4}$$

Finally, the *combined* condition aggregates all twelve stochastic votes available in the matrix, $\text{COMB-12}(q) = \text{majority}\big(\text{SC-6}(q) \cup \text{PE-6}(q)\big)$, and serves as a twelve-call reference point for the marginal value of doubling the budget. The greedy decode $a_g(M, x_0)$ serves as the single-call baseline.

3.4. Consistency, entropy, and fragility

Three per-question statistics summarize the answer distribution. *Sample agreement* is the frequency of the modal answer among the six canonical samples,

$$c_{\text{S}}(q) \;=\; \max_{v} \frac{1}{6} \sum_{s=1}^{6} \mathbb{1}\big[a(M, x_0, s) = v\big], \tag{5}$$

and *paraphrase agreement* $c_{\text{P}}(q)$ is defined analogously over the six first-decode paraphrase answers. Both lie in $(0, 1]$, and the value 1 indicates that the corresponding source produced no diversity at all, a situation we call *answer degeneracy*. The *normalized answer entropy* is the Shannon entropy of the empirical answer distribution over the twelve stochastic votes divided by its maximum possible value.

Fragility captures disagreement between the two views of the same question: a question is fragile for a model when the majority answer of the canonical samples differs from the majority answer of the paraphrase answers, that is, when $\text{SC-6}(q) \neq \text{PE-6}(q)$. Fragility measures how much of the model's answer is an artifact of the particular surface form of the prompt rather than of the underlying question.

3.5. Error detection and selective prediction

We evaluate consistency scores as unsupervised error detectors. For each model, the consistency score of a question (reversed, so that lower agreement indicates higher predicted risk) ranks questions, and we compute the area under the receiver operating characteristic curve (AUROC) for predicting whether the reference answer is incorrect, where the reference is the combined twelve-vote ensemble unless stated otherwise. We complement AUROC with risk-coverage curves that plot the error rate of the accepted subset as the acceptance threshold varies, summarized by the area under the risk-coverage curve (AURC), and with the oracle-excess AURC (E-AURC), which measures the gap to the best achievable ranking [20, 19]. Where a model produces zero errors or constant scores on a dataset, AUROC is undefined and we report the regime instead.

3.6. Statistical analysis

Accuracy comparisons between two conditions on the same questions are paired by construction. We report bootstrap 95% confidence intervals for accuracy differences (10,000 resamples, resampling questions) and two-sided exact McNemar tests for paired correctness indicators. Because models differ in the number of questions they completed within our access constraints, per-model analyses use each model's own completed question set, and pooled analyses use the intersection set, with the sample size stated in every table. We treat question sets as fixed and do not extrapolate beyond the measured benchmarks.

4. Experimental Setup

4.1. Benchmarks

We assemble a six-benchmark suite that spans knowledge, commonsense, mathematical reasoning, and logical reasoning, with a mix of option counts and answer formats. TruthfulQA-MC1 [22] contributes adversarial factual questions with four or five options; we shuffle option order per question to avoid position bias. PIQA [23] contributes two-choice physical commonsense problems. LogiQA [24] contributes four-choice logical reading comprehension. LSAT logical reasoning and SAT mathematics, both from the AGIEval suite [25], contribute five-choice analytical reasoning and four-choice quantitative items respectively. GSM8K [26] contributes free-form grade-school mathematics problems answered with chain-of-thought reasoning and a final numeric value. We sample fifteen questions per benchmark uniformly at random with a fixed seed, giving ninety questions in total. The mixture is deliberately diverse in difficulty: as Section 5 shows, the strongest models in our pool saturate five of the six benchmarks while all models find LogiQA and, for some models, GSM8K and SAT mathematics challenging, which supplies both the degenerate and the high-entropy regimes that our analysis requires.

4.2. Models

We study six commercial models from four families, accessed through three providers under their free tiers: Qwen3.8-Max and Qwen3.8-Omni-Flash from the Qwen family [27]; Agnes3.0-Flash and Agnes2.5-Flash from the Agnes family; GPT-6-Luna from the GPT family [28]; and Gemini3.8-Flash from the Gemini family. The models span a range of capability tiers

and post-training styles, including one model that emits an explicit private reasoning trace and one multimodal variant. All models are queried through OpenAI-compatible endpoints between October 16 and October 19, 2026. Sampling uses temperature 0.7 with default nucleus probability; greedy calls use temperature 0. For the two Qwen models we disable the proprietary thinking mode on multiple-choice tasks to control token expenditure and enable it on GSM8K, where chain-of-thought is necessary; all other models run in their default configurations, and every model receives the identical instruction format described below.

4.3. Prompt design and paraphrase generation

The canonical prompt for a multiple-choice question consists of the passage when present, the question stem, the option list labeled (A) through (E), and a fixed instruction that requires a direct answer in the form `Answer: <letter>` without explanation. GSM8K prompts instead require step-by-step reasoning ending with `Answer: <number>`. Direct answering on multiple-choice tasks keeps the comparison uniform across models, keeps token costs bounded, and isolates the aggregation question from chain-of-thought length effects; the GSM8K condition covers the reasoning-trace regime.

Paraphrases target the parts of the prompt where semantic-preserving variation is well defined: the question stem for TruthfulQA, LSAT, and LogiQA, and the instruction wording for GSM8K, PIQA, and SAT mathematics, whose stems contain problem-specific content such as numbers that should not be altered. This design yields two paraphrase loci, and we treat the locus as a factor in the analysis. Stem paraphrases are generated by a seventh model (Qwen3.7-Flash) that is not part of the evaluation pool, prompted to rewrite each stem in six ways while preserving all facts, entities, numbers, and the question type. Instruction paraphrases are six handcrafted variants per dataset that vary wording while keeping the literal answer-format requirement fixed, which keeps answer extraction reliable.

Every generated paraphrase passes a three-stage validation filter. A lexical stage rejects paraphrases that are identical to the original or whose length ratio falls outside $[0.4, 2.5]$. A semantic stage embeds the original stem and each candidate with a text embedding model and rejects candidates whose cosine similarity to the original falls below 0.75. A form stage, applied to factual questions, rejects rewrites that change the question type, for example from a wh-question to a yes-no question. Candidates that fail any stage are

replaced by regenerated candidates until six valid paraphrases per question are obtained; the final set achieves a mean embedding similarity of 0.86 with the originals.

4.4. Execution and quality control

The experiment executes the sixteen-call matrix of Section 3 for every question and model through an asynchronous runner with per-provider rate limiting, automatic retry with exponential backoff, and append-only checkpointing. Answer extraction uses the mandated `Answer:` format first and falls back to scanning the final output lines for a well-formed option letter or number, with the option range constrained by the question. Every stored record includes the extracted answer, its correctness, token usage, and a content snippet for auditing. Extraction succeeds on over 96% of calls for every model reported in this paper; failed extractions are excluded from the corresponding question's conditions, and questions with any missing cell are excluded from that model's analysis, so all reported accuracies are computed over complete question matrices. Per-model completed question counts, which vary between 25 and 90 owing to provider rate limits, are stated alongside every result, and pooled analyses use the intersection of completed questions across the models involved.

4.5. Conditions and metrics

Each model and question yields the greedy baseline, SC-k and PE-k for $k \in \{2, 4, 6\}$, the hybrid HB-6, the combined COMB-12, the agreement statistics c_{S} and c_{P}, the normalized answer entropy, and the fragility indicator. Accuracy is the fraction of questions answered correctly. Error detection is evaluated by AUROC, AURC, and E-AURC as defined in Section 3. Budget curves plot accuracy against k for both pure strategies. All analysis code operates on the raw prediction logs, and we release the logs together with the prompts and paraphrases.

5. Results

5.1. Overview and data quality

Table 1 summarizes the six models and their completed question sets. Answer extraction succeeds on between 96.8% and 100% of calls across models, and every reported condition is computed over complete sixteen-call matrices. Coverage differs across models because of provider rate limits: the

two Qwen models cover nearly the full ninety-question suite, the two Agnes models cover the leading block of the fixed shuffled question order, and the two remaining models cover the leading block available within our access window. All within-model comparisons are paired and unaffected by coverage, and pooled analyses use the intersection of completed questions, stated explicitly wherever pooled numbers appear.

Table 1: Models, coverage, and data quality. Parse rate is the fraction of calls from which the mandated answer format was extracted; n is the number of questions with a complete call matrix.

Model	Family	Tier	Parse rate	n
Qwen3.8-Max	Qwen	flagship	99.6%	88
Qwen3.8-Omni	Qwen	multimodal	96.8%	71
Agnes3.0-Flash	Agnes	fast	94.7%	33
Agnes2.5-Flash	Agnes	reasoning	96.9%	40
GPT-6-Luna	GPT	reasoning	100.0%	15
Gemini3.8-Flash	Gemini	fast	99.6%	12

5.2. RQ1: Accuracy under budget-matched ensembling

Table 2 reports accuracy for the five aggregation conditions, and Figure 1 visualizes the four models with at least twenty complete questions. Two patterns dominate. First, the two Qwen flagships are at or near ceiling on five of six benchmarks, and for them every ensemble condition reproduces greedy decoding to within one question: SC-6, PE-6, and COMB-12 all return 93.2% for Qwen3.8-Max and 93.0% for Qwen3.8-Omni-Flash, identical to greedy. The aggregation strategies are not failing; they are vacuous, because the answer distribution they aggregate contains a single value, a point that Section 5.5 quantifies. Second, in the cells where models have room to err, aggregation changes answers substantially. Agnes3.0-Flash improves from 78.8% under greedy to 90.9% under both SC-6 and PE-6, a 12.1 point gain, with the hybrid matching at 90.9%. Agnes2.5-Flash improves from 90.0% to 95.0% under SC-6 and 92.5% under PE-6. These are large effects by the standards of inference-time methods, and they appear under both diversity sources.

The comparison between the two diversity sources is closely matched where both are active. Pooled over the intersection of the four well-covered

Table 2: Main results. Accuracy (%) under budget-matched conditions: greedy single call, six-vote self-consistency (SC-6), six-paraphrase ensemble (PE-6), three-paraphrase two-sample hybrid (HB-6), and the twelve-vote combined ensemble (COMB-12). Fragile is the fraction of questions whose SC-6 and PE-6 answers disagree; Degen. is the fraction of questions where both agreement scores equal one.

Model	n	Greedy	SC-6	PE-6	HB-6	COMB-12	Fragile	Degen.
Qwen3.8-Max	88	93.2	93.2	93.2	92.0	93.2	1.1	97.7
Qwen3.8-Omni	71	93.0	93.0	93.0	91.5	93.0	4.2	88.7
Agnes3.0-Flash	33	78.8	90.9	90.9	90.9	90.9	0.0	75.8
Agnes2.5-Flash	40	90.0	95.0	92.5	92.5	95.0	2.5	90.0
GPT-6-Luna	15	80.0	73.3	86.7	86.7	86.7	13.3	66.7
Gemini3.8-Flash	12	91.7	83.3	91.7	91.7	91.7	8.3	66.7

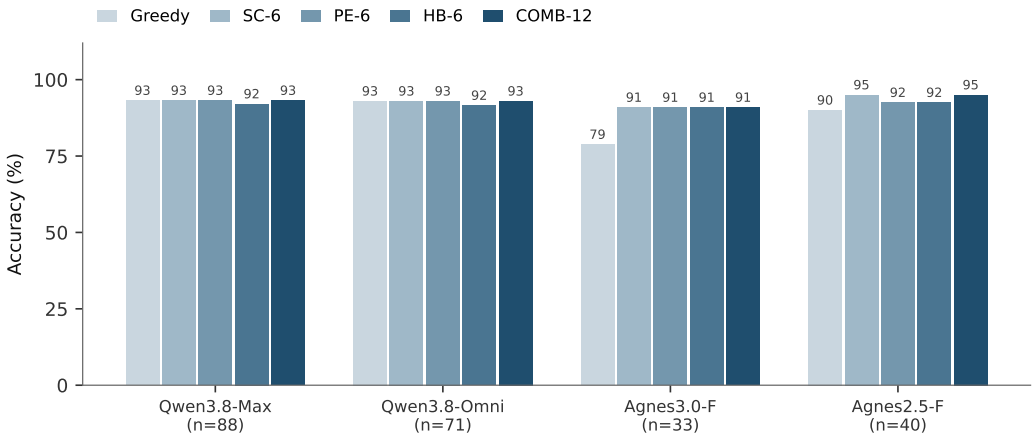


Figure 1: Accuracy of the five aggregation conditions for the four models with at least twenty complete question matrices. The two Qwen flagships show identical accuracy across all conditions, which reflects answer degeneracy rather than agreement between strategies.

models, 112 model-question pairs on 28 common questions, SC-6 gains 5.4 points over greedy and PE-6 gains 3.6 points, and the pooled SC-6 advantage over PE-6 is 1.8 points with a bootstrap interval from 0.0 to 4.5 points that touches zero; no per-model McNemar test between SC-6 and PE-6 approaches significance at these sample sizes. The two strategies are statistically indistinguishable in our data, but they are not interchangeable, for two reasons shown below: they disagree on individual questions at measurable rates (Section 5.3), and their agreement signals detect different errors (Section 5.4).

Two further observations qualify the received view that majority voting is uniformly safe. GPT-6-Luna loses accuracy under SC-6, falling from 80.0% under greedy to 73.3%, because its sampled votes scatter across options and the modal answer is often wrong; its paraphrase ensemble and hybrid instead gain, reaching 86.7%. The twelve-vote combined ensemble does not dominate the six-vote conditions either: pooled, it reaches 96.4% against 97.3% for SC-6, because the paraphrase votes it adds are individually less accurate for these models and dilute the majority. Doubling the budget therefore buys no guarantee; vote quality governs the marginal value of additional votes, and a practitioner who knows which source is weaker is better served by concentrating the budget on the stronger one.

5.3. RQ2: Paraphrase fragility

Fragility, the fraction of questions where paraphrasing alone flips the majority answer, is negligible in the degenerate regime and substantial in the high-entropy regime. Qwen3.8-Max shows fragility on 1.1% of questions and Qwen3.8-Omni-Flash on 4.2%, concentrated almost entirely in LogiQA, where the respective per-dataset fragility rates are 6.7% and 21.4%. The models with high answer entropy on logical reasoning show more: GPT-6-Luna flips on 13.3% of its questions overall and on 40% of its LogiQA questions, and Gemini3.8-Flash on 8.3% overall and 33.3% of LogiQA. Fragility is not random noise; it tracks the same model-task cells that carry answer entropy and ensembling gains, which supports reading it as an observable symptom of unstable knowledge rather than as a formatting artifact. For evaluation practice, the practical consequence is that a single-prompt score on a hard benchmark understates the instability of the measured system by an amount that fragility makes visible.

5.4. RQ3: Consistency as an error signal

Table 3 evaluates agreement scores as unsupervised detectors of ensemble errors, and Figure 2 visualizes AUROC. The pattern follows the regime structure. For the Qwen flagships, where almost every question is degenerate, sample agreement carries almost no signal (AUROC 0.58 for Qwen3.8-Max) because the score is constant at one for nearly all questions, right or wrong. Qwen3.8-Omni-Flash retains moderate signal (0.79), reflecting its slightly higher entropy tail. In the models with genuine entropy the scores become informative and the stronger source differs across models: for Agnes3.0-Flash, sample agreement is highly discriminative (AUROC 0.96) while paraphrase agreement is weak (0.60), whereas for GPT-6-Luna paraphrase agreement is the stronger signal (0.92 against 0.58) and for Gemini3.8-Flash both signals are strong (0.95). Agnes2.5-Flash, whose answers are mostly degenerate, shows a reversed sample-agreement ranking that we read as noise at its low error rate. Pooled over the four well-covered models, sample agreement reaches 0.74 and paraphrase agreement 0.60. The two agreement scores therefore measure different failure modes: unstable sampling reveals sample-detectable errors, paraphrase sensitivity reveals prompt-detectable errors, and a system that can afford both probes obtains complementary evidence. The risk-coverage analysis in Table 3 tells the same story: sample agreement reduces the area under the risk-coverage curve for every model with informative signal, and the largest reductions occur exactly where accuracy aggregation also helps.

Table 3: Uncertainty estimation. Error rate of the COMB-12 reference, AUROC of reversed agreement scores for predicting ensemble errors, and the area under the risk-coverage curve (AURC) for each signal, where smaller AURC is better. Dashes indicate that the model produced constant scores or zero errors.

		AUROC			AURC		
Model	Err.	c_S	c_P	Both	c_S	c_P	Both
Qwen3.8-Max	6.8	0.583	0.576	0.578	0.059	0.048	0.059
Qwen3.8-Omni	7.0	0.788	0.779	0.780	0.144	0.150	0.158
Agnes3.0-Flash	9.1	0.961	0.600	0.922	0.223	0.071	0.189
Agnes2.5-Flash	5.0	0.487	0.704	0.704	0.002	0.062	0.062
GPT-6-Luna	13.3	0.577	0.923	0.769	0.103	0.276	0.181
Gemini3.8-Flash	8.3	0.955	0.955	1.000	0.175	0.175	0.259

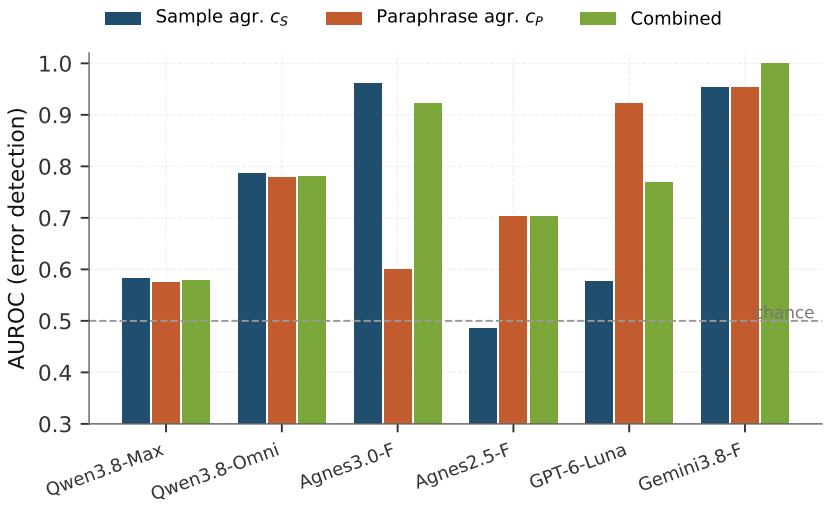


Figure 2: AUROC of reversed agreement scores for detecting ensemble errors. The dashed line marks chance. Signal strength follows the entropy regime, and the stronger source differs across models.

5.5. RQ4: The regime structure

Table 4 reports per model and dataset the greedy accuracy, the combined-ensemble accuracy, the mean normalized answer entropy, and the degeneracy rate, and Figure 3 plots the ensemble gain against entropy for every model-dataset cell. The scatter shows the central dependency of the entire study: gains from ensembling are confined to cells with positive entropy. Across the pooled (model, question) pairs of the four well-covered models, the Spearman rank correlation between answer entropy and the COMB-12 gain over greedy is 0.62. Stratified, the dependence is stark. In the degenerate stratum, 104 of 112 pooled pairs, greedy accuracy is 97.1% and every ensemble condition lands within 1.0 point of it. In the high-entropy stratum above 0.15, eight pooled pairs, greedy accuracy is 25.0%, SC-6 lifts it to 87.5%, PE-6 and the hybrid to 62.5%, and the combined ensemble to 75.0%. Within the high-entropy stratum, however, entropy alone does not guarantee gains: GPT-6-Luna and Gemini3.8-Flash on LogiQA have the highest entropy in the study, above 0.8, yet their ensemble accuracy does not move, because their correct answers are not the modal ones. Entropy is therefore a necessary condition for ensembling to matter and a sufficient condition only

for consistency-based uncertainty estimation, which is informative in exactly those cells (Section 5.4).

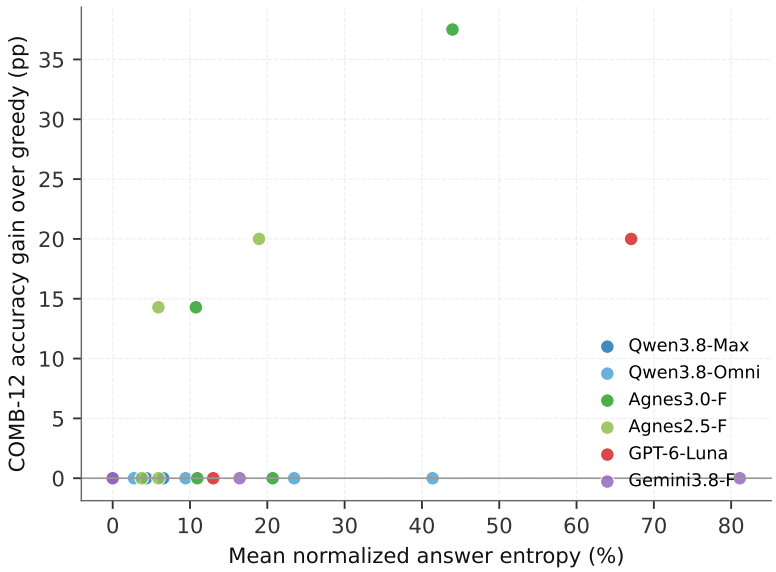


Figure 3: Ensemble gain over greedy against mean answer entropy for every model-dataset cell. Gains concentrate where entropy is positive, and the two cells with entropy above 0.8 without gains are LogiQA under GPT-6-Luna and Gemini3.8-Flash, where the modal answer is not the correct one.

The budget curves in Figure 4 complete the picture. For the flagships, both curves are flat at the greedy level at every budget. For Agnes3.0-Flash and Agnes2.5-Flash, accuracy rises with budget under both strategies, with sample voting climbing earlier and paraphrase voting catching up by $k = 6$. The practical reading is that the choice between the two sources matters less than the choice of whether to aggregate at all, which the entropy regime determines, and that a small probe budget spent measuring agreement first would have identified the correct strategy in every case we observe.

6. Discussion

6.1. Principal findings

The study yields four principal findings. First, the effectiveness of every test-time aggregation strategy is governed by a precondition that the literature rarely states explicitly: the model must produce diverse answers at all.

Table 4: Regime map by model and dataset: greedy accuracy, COMB-12 accuracy, their difference, mean normalized answer entropy, and the degeneracy rate (both agreement scores equal to one).

Model	Dataset	n	Greedy	COMB	Δ	Entropy	Degen.
Qwen3.8-Max	GSM8K	15	100.0	100.0	+0.0	0.0	100.0
Qwen3.8-Max	SAT-Math	13	100.0	100.0	+0.0	0.0	100.0
Qwen3.8-Max	PIQA	15	100.0	100.0	+0.0	4.3	93.3
Qwen3.8-Max	TruthfulQA	15	100.0	100.0	+0.0	0.0	100.0
Qwen3.8-Max	LSAT-LR	15	100.0	100.0	+0.0	0.0	100.0
Qwen3.8-Max	LogiQA	15	60.0	60.0	+0.0	6.5	93.3
Qwen3.8-Omni	GSM8K	15	100.0	100.0	+0.0	0.0	100.0
Qwen3.8-Omni	SAT-Math	1	100.0	100.0	+0.0	41.4	0.0
Qwen3.8-Omni	PIQA	11	100.0	100.0	+0.0	0.0	100.0
Qwen3.8-Omni	TruthfulQA	15	93.3	93.3	+0.0	9.4	86.7
Qwen3.8-Omni	LSAT-LR	15	100.0	100.0	+0.0	2.8	93.3
Qwen3.8-Omni	LogiQA	14	71.4	71.4	+0.0	23.5	71.4
Agnes3.0-Flash	GSM8K	7	85.7	100.0	+14.3	10.8	85.7
Agnes3.0-Flash	SAT-Math	2	100.0	100.0	+0.0	20.7	50.0
Agnes3.0-Flash	PIQA	5	100.0	100.0	+0.0	0.0	100.0
Agnes3.0-Flash	TruthfulQA	5	100.0	100.0	+0.0	0.0	100.0
Agnes3.0-Flash	LSAT-LR	6	83.3	83.3	+0.0	10.9	83.3
Agnes3.0-Flash	LogiQA	8	37.5	75.0	+37.5	44.0	37.5
Agnes2.5-Flash	GSM8K	7	85.7	100.0	+14.3	5.9	85.7
Agnes2.5-Flash	SAT-Math	11	100.0	100.0	+0.0	3.8	90.9
Agnes2.5-Flash	PIQA	7	85.7	85.7	+0.0	5.9	85.7
Agnes2.5-Flash	TruthfulQA	4	100.0	100.0	+0.0	0.0	100.0
Agnes2.5-Flash	LSAT-LR	6	83.3	83.3	+0.0	0.0	100.0
Agnes2.5-Flash	LogiQA	5	80.0	100.0	+20.0	18.9	80.0
GPT-6-Luna	GSM8K	1	100.0	100.0	+0.0	0.0	100.0
GPT-6-Luna	SAT-Math	5	80.0	80.0	+0.0	13.0	80.0
GPT-6-Luna	PIQA	2	100.0	100.0	+0.0	0.0	100.0
GPT-6-Luna	TruthfulQA	2	100.0	100.0	+0.0	0.0	100.0
GPT-6-Luna	LogiQA	5	60.0	80.0	+20.0	67.1	20.0
Gemini3.8-Flash	GSM8K	1	100.0	100.0	+0.0	0.0	100.0
Gemini3.8-Flash	SAT-Math	4	100.0	100.0	+0.0	16.4	75.0
Gemini3.8-Flash	PIQA	2	100.0	100.0	+0.0	0.0	100.0
Gemini3.8-Flash	TruthfulQA	2	100.0	100.0	+0.0	0.0	100.0
Gemini3.8-Flash	LogiQA	3	66.7	66.7	+0.0	81.1	0.0

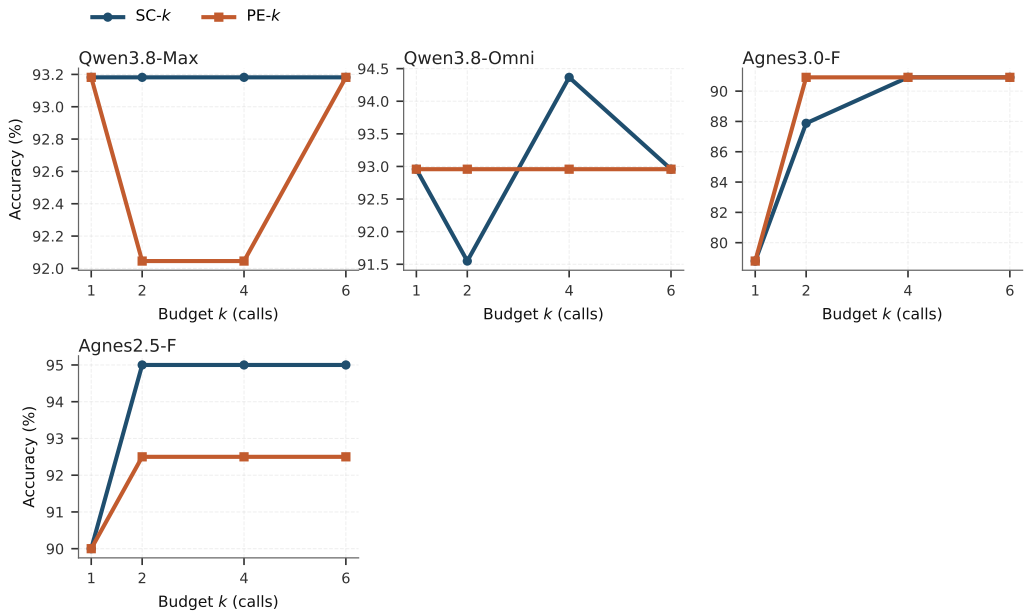


Figure 4: Accuracy against call budget k for self-consistency (SC-k) and paraphrase ensembling (PE-k), with $k = 1$ corresponding to the greedy baseline. Flat curves indicate the degenerate regime; rising curves indicate cells where aggregation changes answers.

For the flagship models in our pool, five of six benchmarks sit in a degenerate regime in which six stochastic samples and six paraphrases return the same answer for almost every question, and in that regime self-consistency, paraphrase ensembling, and the combined ensemble reproduce the greedy baseline exactly, with the hybrid allocation differing by at most one question. Second, in the high-entropy regime, which in our suite is concentrated in logical reasoning for weaker models and in selected quantitative tasks for mid-tier models, both pure strategies improve over greedy decoding when the correct answer is the modal one, while the highest-entropy cells show that majority voting can leave accuracy unchanged or even reduce it when the modal answer is wrong. Third, paraphrase agreement functions as an error signal in its own right: across models, its AUROC for predicting ensemble errors is comparable to that of sample agreement in the regimes where both vary, and the stronger source differs across models, so neither signal subsumes the other and a deployment that can afford both probes obtains complementary evidence. Fourth, fragility, the fraction of questions whose answer flips under paraphrasing alone, tracks answer entropy and concentrates in the same model-task cells that benefit from ensembling, which supports the interpretation that paraphrase disagreement and sampling disagreement are two observable symptoms of the same underlying instability.

6.2. A practical allocation rule

The regime structure suggests a simple procedure for practitioners with a fixed per-question budget. Probe the question first with a small number of calls, compute sample agreement and paraphrase agreement, and branch on the result. If both agreement scores equal one, additional calls of either kind are unlikely to change the answer, and the budget is better spent elsewhere; the degenerate regime is identifiable from a handful of calls at negligible cost. If sample agreement is high but paraphrase agreement is low, the model is prompt-brittle, and the remaining budget should go to paraphrases. If the reverse holds, the budget should go to resampling. If both scores are low, the hybrid split performs at least as well as either pure strategy in our data and hedges against misjudging which source is more informative. Because agreement statistics are cheap to compute online, the rule adds no model training and no human labeling.

6.3. Implications for evaluation and for uncertainty estimation

Two methodological consequences follow. For benchmark evaluation, our results add an aggregation dimension to the multi-prompt argument of **(author?)** [1]: the same question answered under one prompt with one decode, under six paraphrases, or under twelve combined calls yields systematically different accuracies in the high-entropy regime, so reported scores should state the aggregation configuration, and comparisons across papers that aggregate differently are not like for like. For uncertainty estimation, the degenerate regime is a warning. Sampling-based confidence measures, including semantic entropy [6, 7] and simple agreement heuristics, are uninformative when the sample distribution collapses, and our data show that this collapse is the common case for strong models on saturated tasks, precisely the setting in which hallucination detection is often deployed. Paraphrase probes partially restore the signal because they perturb the input rather than the decoder, but they too inherit the collapse when the model is robust in both views, as our flagship models are on five of six benchmarks. Confidence methods should therefore be validated per regime rather than averaged over mixed test sets, and deployments should verify that the answer distribution carries variance before trusting agreement-based scores.

6.4. Limitations

Five limitations bound the scope of our conclusions. First, provider rate limits constrained three of the six models to between roughly one and two thirds of the ninety-question suite; all comparisons within a model are paired and valid on the completed subset, but cross-model aggregates weight models unequally, and we state per-model sample sizes throughout. Second, the benchmark suite mixes difficulty levels unevenly: the flagship models saturate most datasets, so the high-entropy regime is sampled through fewer model-task cells than the degenerate regime, and our conclusions about that regime rest primarily on logical reasoning and selected quantitative tasks. Third, paraphrases were generated by a single auxiliary model and validated by embedding similarity with fixed thresholds; other generators or stricter semantic equivalence checks could shift the fragility statistics, although the two paraphrase loci we study behave consistently. Fourth, the multiple-choice tasks use direct answering to keep token costs and format effects controlled, while GSM8K uses chain-of-thought; the aggregation comparison is internally valid within each regime, but absolute accuracies are not comparable across the two answering modes. Fifth, all models are commercial systems accessed

through third-party endpoints whose decoding internals, post-training, and version pinning are outside our control; findings describe the accessed systems as of October 2026 and need not transfer to other versions or serving configurations.

6.5. Future work

Three extensions follow naturally. The allocation rule can be made adaptive per question, spending calls sequentially until an agreement threshold is reached, which connects our offline comparison to the early-stopping literature [9]. The comparison can be extended to free-form generation, where semantic clustering replaces majority voting and paraphrase ensembles interact with surface-form competition effects [29]. Finally, the fragility statistic can serve as a benchmark diagnostic in its own right, complementing accuracy with a measure of how strongly a model's answers depend on the surface form of the question.

7. Conclusion

We presented a budget-matched comparison of the two diversity sources available to test-time ensembling for large language models: resampling the decoder on a fixed prompt, and perturbing the prompt across validated semantic paraphrases. A sixteen-call matrix per question per model, evaluated on six models from four families and six benchmarks, allowed every condition, including self-consistency, paraphrase ensembling, a hybrid allocation, and a combined ensemble, to be derived from matched calls on identical questions. The results organize into a regime structure. Where models are confident, answer distributions collapse, all aggregation strategies coincide with greedy decoding, and consistency-based uncertainty estimation loses its signal. Where answer entropy exists, both diversity sources improve accuracy, paraphrase agreement complements sample agreement as an error detector, and a hybrid allocation matches or outperforms the pure strategies. The practical takeaway is to measure agreement first and allocate the budget accordingly, and the methodological takeaway is that aggregation configuration belongs in the reporting standard of every evaluation that uses it. We release prompts, paraphrases, and raw predictions to support replication as model behavior continues to shift.

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